This discovery, that all wormholes require exotic material to hold them open, triggered much theoretical research during 1988–92. “Do the laws of physics permit exotic material to exist, and if so, under what circumstances?” This was the central issue.

A key to the answer had already been provided in the 1970s by Stephen Hawking. In 1970, when proving that the surface areas of black holes always increase (Chapter 12), Hawking had to assume that there is *no* exotic material near any black hole’s horizon. If exotic material were in the horizon’s vicinity, then Hawking’s proof would fail, his theorem would fail, and the horizon’s surface area could shrink. Hawking didn’t worry much about this possibility, however; it seemed in 1970 a rather safe bet that exotic material cannot exist.

Then, in 1974, came a great surprise: Hawking inferred as a by-product of his discovery of black-hole evaporation (Chapter 12) that *vacuum fluctuations near a hole’s horizon are exotic:* They have negative average energy density as seen by outgoing light beams near the hole’s horizon. In fact, it is this exotic property of the vacuum fluctuations that permits the hole’s horizon to shrink as the hole evaporates, in violation of Hawking’s area-increase theorem. Because exotic material is so important for physics, I shall explain this in greater detail:

Recall the origin and nature of vacuum fluctuations, as discussed in Box 12.4: When one tries to remove all electric and magnetic fields from some region of space, that is, when one tries to create a perfect vacuum, there always remain a plethora of random, unpredictable electromagnetic oscillations-oscillations caused by a tug-of-war between the fields in adjacent regions of space. The fields “here” borrow energy from the fields “there,” leaving the fields there with a deficit of energy, that is, leaving them momentarily with negative energy. The fields there then quickly grab the energy back and with it a little excess, driving their energy momentarily positive, and so it goes, onward and onward.

Under normal circumstances on Earth, the average energy of these vacuum fluctuations is zero. They spend equal amounts of time with energy deficits and energy excesses, and the average of deficit and excess vanishes. Not so near the horizon of an evaporating black hole, Hawking’s 1974 calculations suggested. Near a horizon the average energy must be negative,